

Iteration 1

[0]	[1]	[2]	[3]	[4]	[5]
30	20	60	50	10	40

swap

Pos pos+1

30 > 20 ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
20	30	60	50	10	40

Pos pos+1

30 > 60 ? no

[0]	[1]	[2]	[3]	[4]	[5]
20	30	60	50	10	40

swap

Pos pos+1

60 > 50 ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	60	10	40

Pos pos+1

60 > 10 ? Yes

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	10	60	40

swap

Pos pos+1

60 > 40 ? Yes
swap

5 comps.

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	10	40	60

Iteration/cycle/pass : 2

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	10	40	60

Pos pos+1

20 > 30 ? no

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	10	40	60

Pos pos+1

30 > 50 ? no

[0]	[1]	[2]	[3]	[4]	[5]
20	30	50	10	40	60

swap

Pos pos+1

50 > 10 ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
20	30	10	50	40	60

swap

Pos pos+1

50 > 40 ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
20	30	10	40	50	60

Iteration 3

[0]	[1]	[2]	[3]	[4]	[5]
20	30	10	40	50	60

Pos pos+1

20 > 30 ? no

1st it → 4th index
→ 2nd it → 3rd index
3rd it → 2nd "
4th it → 1st ind
5th it → 0th.

3 comps

Pos pos+1

[0]	[1]	[2]	[3]	[4]	[5]
20	30	10	40	50	60

30 > 10 ? Yes
swap

Pos pos+1

[0]	[1]	[2]	[3]	[4]	[5]
20	10	30	40	50	60

30 > 40 ? no

Pos pos+1

[0]	[1]	[2]	[3]	[4]	[5]
20	10	30	40	50	60

Iterations 4

swap

[0]	[1]	[2]	[3]	[4]	[5]
20	10	30	40	50	60

20 > 10 ? Yes
swap

Pos pos+1

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	40	50	60

20 > 30 ? no

Pos pos+1

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	40	50	60

Iteration no : 5

1 comp.

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	40	50	60

Arr[pos] > arr[pos+1]
no

Pos pos+1

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	40	50	60

No. of iterations : n-1 : 5

No. of comparisons : (n-1) + (n-2) + (n-3).... 1
5 + 4 + 3 + 2 + 1 = 15

Best Case Time Complexity : $O(n)$
Worst/ Avg case : $O(n^2)$

$O(n^2)$

Space comp :
Aux space : $O(1)$ constant